

MANTIS AI

Enterprise Data Security

Comprehensive Security Documentation



MANTIS

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Municipal Financial Intelligence Platform

EXECUTIVE SUMMARY

Mantis AI implements enterprise-grade security measures specifically designed for sensitive municipal financial data. This comprehensive security framework ensures that all uploaded documents are processed with bank-level protection standards while maintaining complete data confidentiality and regulatory compliance.

The security architecture encompasses memory-only processing, multi-layer validation, comprehensive audit trails, and automatic session management to provide maximum protection for sensitive government financial information.

CORE SECURITY FEATURES

Memory-Only Processing

- * **Zero Disk Storage**

Files are processed entirely in memory and never written to permanent storage

- * **Temporary Processing**

All document processing occurs in volatile memory only

- * **Automatic Cleanup**

Memory is automatically cleared when sessions end or timeout

- * **No File Persistence**

No temporary files, caches, or permanent storage of uploaded content

Session Security Management

- * **1-Hour Session Timeout**

Automatic security cleanup after 60 minutes of inactivity

- * **Session Isolation**

Complete separation between different user sessions

- * **Force Cleanup**

Immediate data removal when sessions expire

- * **User-Controlled Clearing**

Manual security cleanup options available at any time

File Validation & Sanitization

- * **Comprehensive Type Validation**

Only PDF and CSV files accepted, all others rejected

- * **File Size Limits**

50MB maximum file size to prevent system abuse

- * **Content Sanitization**

Automatic removal of malicious patterns and scripts

- * **SQL Injection Protection**

Advanced filtering to prevent database attacks

- * **Hash Verification**

File integrity verification using SHA-256 hashing

SECURITY ARCHITECTURE

Multi-Layer Protection System

Mantis implements a 5-layer security validation system:

- * Layer 1: File type and size validation
- * Layer 2: Content sanitization and malware filtering
- * Layer 3: Memory-only processing enforcement
- * Layer 4: Session timeout and isolation
- * Layer 5: Comprehensive audit logging and monitoring

Secure File Processing Flow

Every uploaded file follows this secure processing workflow:

- * 1. File Reception: Initial security validation
- * 2. Type & Size Check: Comprehensive validation against security policies
- * 3. Content Sanitization: Removal of potentially harmful content
- * 4. Memory Processing: Secure in-memory data extraction
- * 5. Session Storage: Temporary storage in isolated session state
- * 6. AI Analysis: Secure processing with full context protection
- * 7. Automatic Cleanup: Secure data removal and memory clearing

DATA PROTECTION MEASURES

Content Security

* Text Length Limits

Maximum 1MB of text content to prevent system overload

* DataFrame Size Restrictions

Maximum 100,000 rows for CSV data processing

* Column Name Sanitization

Automatic cleaning of potentially dangerous column names

* Pattern Filtering

Removal of HTML/XML tags and script patterns

Access Control & Monitoring

* Complete Audit Trail

Every file operation logged with timestamp and details

* Security Event Tracking

Comprehensive logging of validation, processing, and errors

* File Integrity Monitoring

Hash-based verification of file content integrity

* Real-Time Security Status

Live monitoring of active security measures

COMPLIANCE & STANDARDS

Government Data Protection

Mantis security measures are designed to meet or exceed government data handling requirements for sensitive financial information:

- * Zero-persistence data handling for maximum confidentiality
- * Complete audit trails for regulatory compliance
- * Session-based security isolation
- * Enterprise-grade access logging
- * Automatic data lifecycle management

Security Best Practices Implementation

* Defense in Depth

Multiple security layers provide comprehensive protection

* Principle of Least Privilege

Minimal data retention and access requirements

* Fail-Safe Defaults

Secure-by-default configuration with automatic protections

* Complete Separation

Isolation between processing, storage, and transmission

TECHNICAL IMPLEMENTATION

SecureFileHandler Class

The core security implementation includes:

- * `validate_file_security()`: Comprehensive file validation
- * `process_pdf_securely()`: Memory-only PDF processing
- * `process_csv_securely()`: Secure DataFrame creation
- * `sanitize_text_content()`: Advanced content filtering
- * `sanitize_dataframe()`: Data structure security validation
- * `enforce_session_timeout()`: Automatic cleanup enforcement
- * `clear_file_data()`: Manual security clearing
- * `log_file_access()`: Comprehensive audit logging

Security Event Logging

All security events are logged with complete details:

- * `VALIDATED` - File passed security checks
- * `PROCESSING_PDF/CSV` - File processing initiated
- * `PROCESSED_PDF/CSV` - File successfully processed
- * `REJECTED_SIZE/TYPE` - File rejected for security reasons
- * `ERROR_PDF/CSV` - Processing errors logged
- * `CRITICAL_ERROR` - System errors tracked
- * `CLEARED` - Data removal confirmation

USER SECURITY CONTROLS

Security Information Panel

Users have complete visibility into security measures:

- * Real-time security status display
- * Active security feature confirmation
- * Session timeout countdown
- * File size and type limit display
- * Current file count in secure memory

Manual Security Operations

* Immediate Data Clearing

Users can manually clear all uploaded files instantly

* Security Status Monitoring

Real-time view of active protections

* File Processing Confirmation

Visual confirmation of secure processing

* Security Hash Display

File integrity verification visible to users

SECURITY VERIFICATION

How Users Can Verify Security

The following indicators confirm enterprise security is active:

- * Security lock icons on all file operations
- * Secure processing confirmations
- * Real-time security status panels
- * Automatic timestamp tracking
- * Manual security cleanup controls
- * File count monitoring in secure memory

Enterprise Security Certification

This security implementation provides:

- * Bank-level data protection standards
- * Government compliance for sensitive financial data
- * Enterprise audit trail requirements
- * Professional security monitoring and controls
- * Zero-risk data persistence policies

All uploaded municipal financial documents are processed with the same security standards used by major financial institutions and government agencies.

SECURITY SUMMARY

Mantis AI provides comprehensive enterprise-grade security for sensitive municipal financial document processing. The multi-layer security architecture ensures complete data protection while maintaining the functionality needed for advanced AI analysis.

Key security principles implemented:

- * Zero persistence: No permanent storage of sensitive data
- * Complete isolation: Session-based security separation
- * Comprehensive monitoring: Full audit trails for compliance
- * Automatic protection: Fail-safe security defaults
- * User control: Manual security management options

This security framework meets the stringent requirements for government financial data handling while providing the advanced AI capabilities needed for modern municipal financial analysis.

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